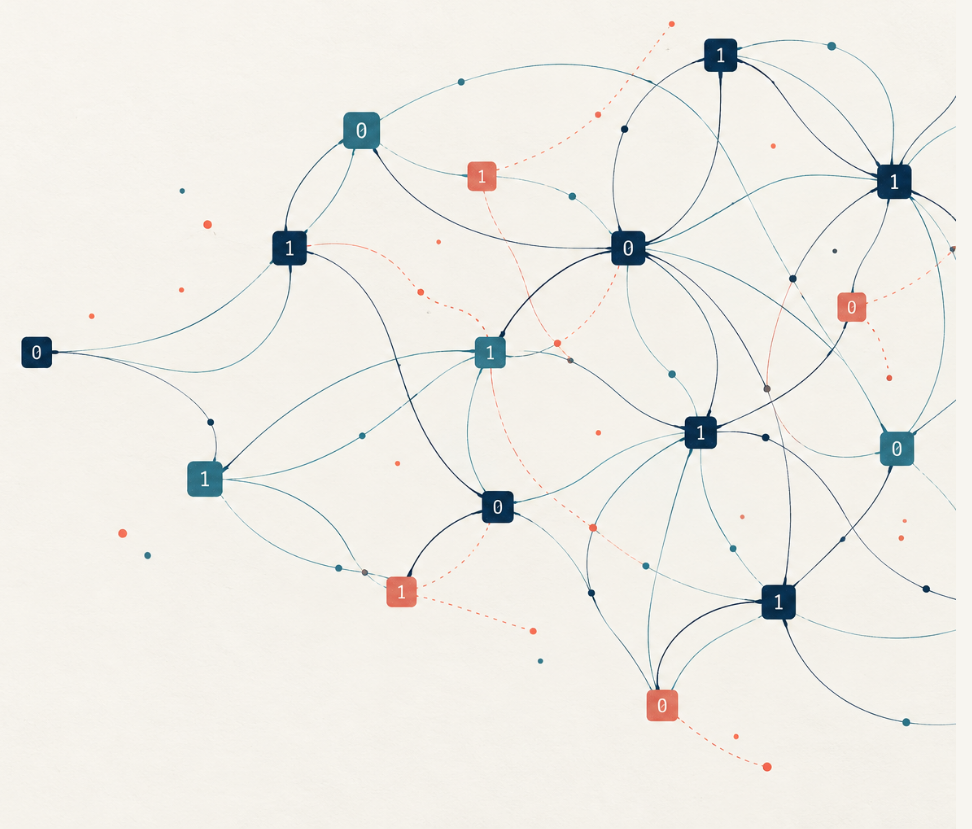


Public-key encryption from sparse noisy equations

An ABW research programme

Assumptions, feasibility and a plan for the first year



Research discussion · 6 October 2026

ABW research proposal

The research objective

Can a hidden sparse structure support useful, concretely analysed encryption?

Research value

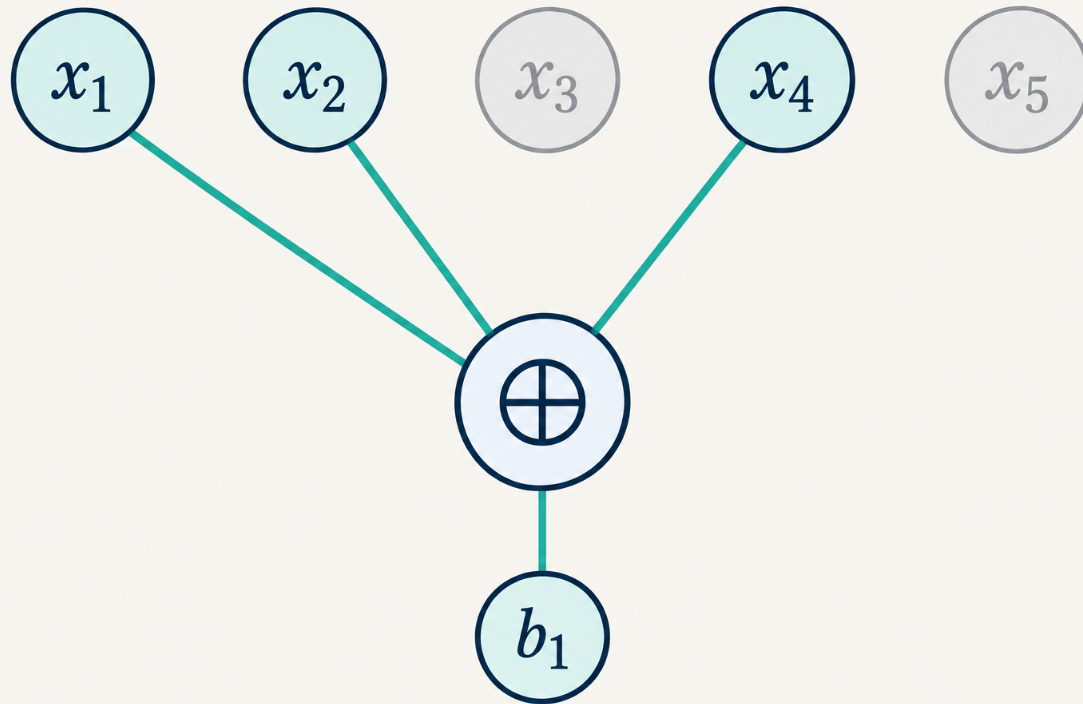
- Different average-case assumptions
- A clear algebraic trapdoor
- Room for sharper security analysis

Evidence still needed

- A defensible parameter set
- Acceptable full construction cost
- Security against active attackers

The initial commitment is a feasibility study with explicit decision points.

Sparse noisy linear equations



$$\mathbf{b} = \mathbf{M}\mathbf{x} + \mathbf{e}$$

$\mathbf{M}$: $m \times n$ binary matrix
Exactly d ones in each row

$\mathbf{x}$: uniform hidden assignment
 $\mathbf{e}_i$: independent $\text{Ber}(\epsilon)$

All arithmetic uses $\mathbb{F}_2$. The diagram shows one noiseless row with $d = 3$.

The hardness task is part of the assumption

Task	Adversary's goal
Search	Recover the sampled assignment x
Approximate search	Recover an assignment close to the planted x
Decision	Distinguish $(M, Mx + e)$ from $(M, \text{uniform})$
Prediction	Predict a specified additional parity of x

Every claim needs the distribution, parameter growth, adversary model and success threshold.

Three routes in ABW

Route	Attraction	Main obstacle
3-LIN alone	Fewest named assumptions	Sampler and amplification cost
d-LIN + DUE	Flexible trapdoor parameters	Additional planted-expansion assumption
DSF + DUE	Nonlinear local construction	Predicate attacks and decoding cost

Proposed starting point: the 3-LIN route, with a separate d-LIN + DUE prototype for comparison.

A short dependency is the trapdoor

Four illustrative 3-sparse rows

1	1	1	0	0	0
1	0	0	1	1	0
0	1	0	1	0	1
0	0	1	0	1	1

$$\mathbf{s}^T \mathbf{M} = \mathbf{0}$$

$$\mathbf{s}^T \mathbf{b} = \mathbf{s}^T \mathbf{e}$$

Each column appears twice.
The assignment x cancels.

Toy algebra only. Security requires a much larger, precisely sampled public key.

The basic ABW bit encryption

Public key: M

Secret: short s , with $s_m = 1$

Encryption

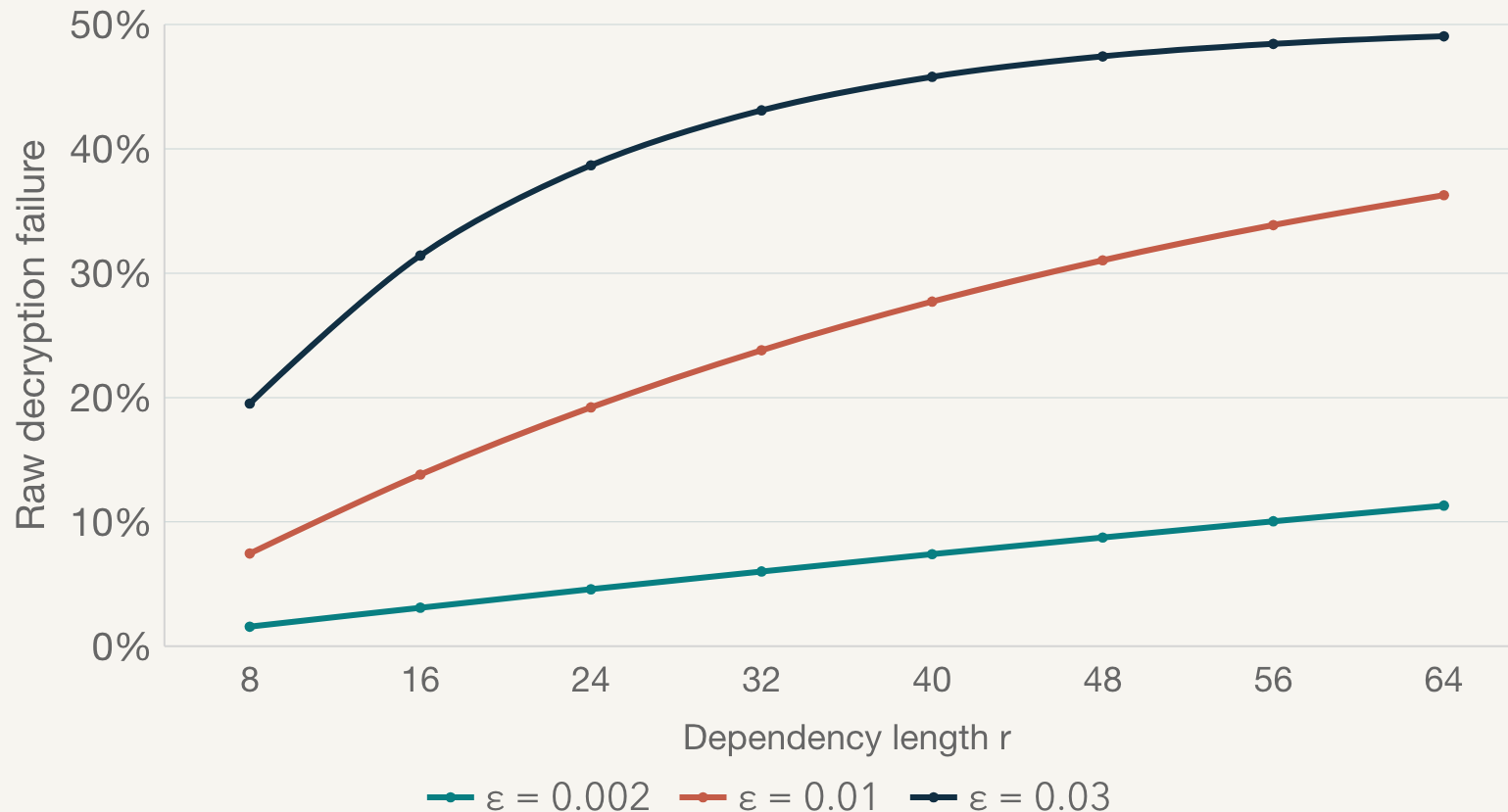
$$c = Mx + e + \mu u_m$$

Decryption

$$s^T c = \mu + s^T e$$

μ is one message bit. Fresh randomness hides it.
The secret relation cancels Mx .

Noise and dependency length control correctness



$$p = [1 - (1 - 2\epsilon)^r] / 2$$

$$p \leq r\epsilon$$

Reliable decoding needs a controlled noise parity.

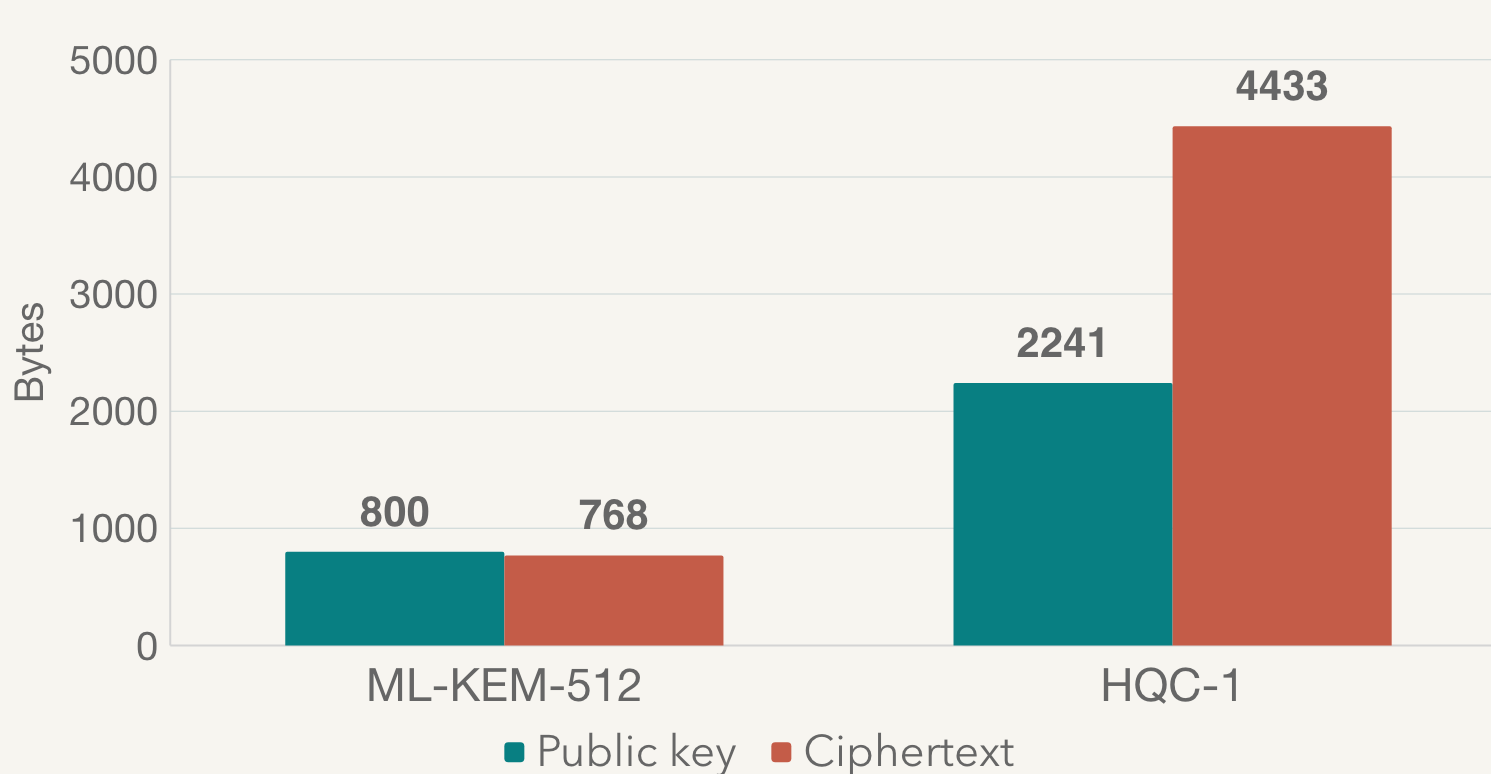
Illustrative calculations. A useful cryptographic parameter choice also needs an attack analysis.

d-LIN, LPN, LWE and Module-LWE

Problem	Coefficients and noise	Security context
Sparse d-LIN	Fixed sparse binary rows Bernoulli output flips	Exact average-case regime matters
Dense LPN	Uniform binary rows Bernoulli output flips	PKE constructions already exist
LWE	Modular coefficients Small modular errors	Lattice reductions in suitable regimes
Module-LWE	Polynomial/module structure Small coefficient errors	Compact structured constructions

Shared equation syntax does not establish a reduction or a security ranking.

Existing KEMs set concrete benchmarks



ABW-derived candidate

Key size: to establish
Ciphertext: to establish

Measure full amplification
and CCA conversion.

Category 1 targets. ML-KEM is standardized. HQC is selected for standardization (specification: 22 Aug 2025).

Attractions and costs of an ABW foundation

Reasons to investigate

Assumption diversification

Simple sparse arithmetic

A trapdoor with clear algebra

Several proof bottlenecks to improve

Costs to quantify

Limited concrete calibration

Large raw bit ciphertexts

Key and ciphertext amplification

CCA and quantum proof obligations

The project needs a measurable advantage: a stronger theorem or a useful concrete trade-off.

Recent work changes the research agenda

01

BKR 2023: public-key sampling matters

Small-subgraph statistics break an alternative planting design.

02

Dao–Jain 2025: parameter boundaries matter

Sparse-LPN attacks apply in particular sample and noise regimes.

03

Hair–Sahai 2026 preprint: new CSP routes

New conjectures and parameters broaden the theoretical programme.

The attacker has two separate targets

Target A: the public matrix

Detect the planted structure.

Find any useful short relation.

Exploit the designated message row.

Target B: noisy ciphertexts

Predict the hidden message bit.

Solve or distinguish noisy equations.

Exploit repeated use of a public key.

Hard random equations do not by themselves hide a planted key.

The first attack suite

Attack family	Question to answer
Short dependencies	Can an attacker find a relation with useful parity bias?
Graph statistics	Do degrees, cycles or small subgraphs expose planting?
Decoding and algebra	Do tailored decoding, SAT or noisy-equation solvers help?
Repeated-key attacks	Does reuse reveal structure absent from a single sample?

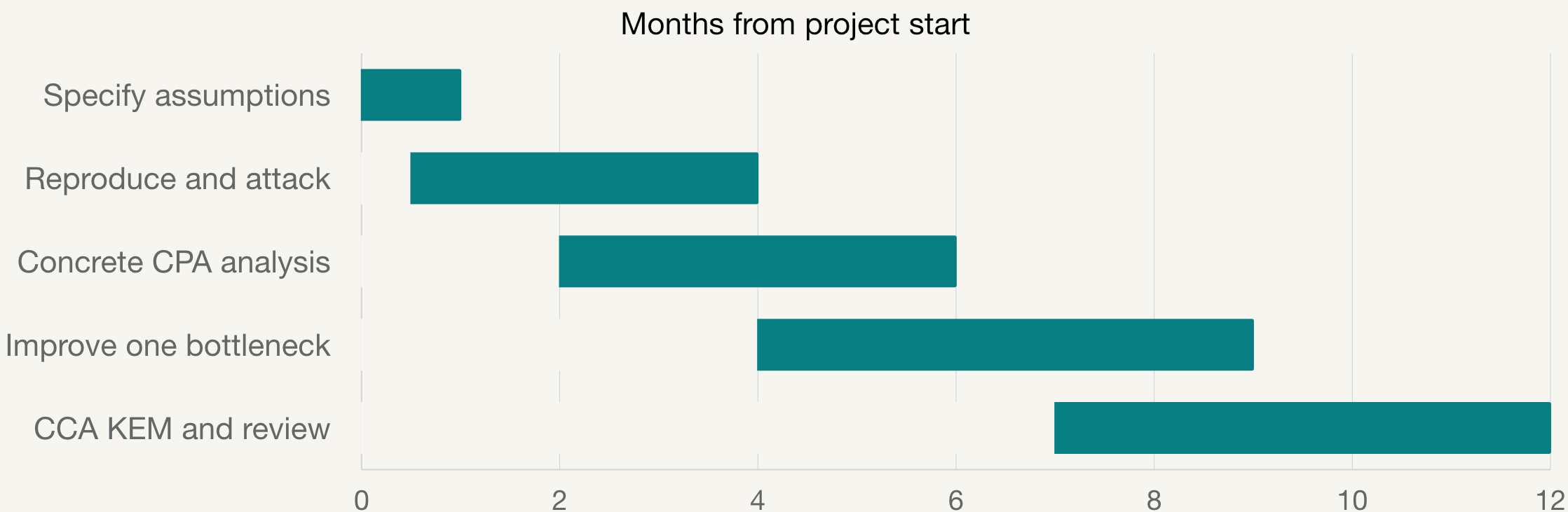
Report successful attacks and their limits. Failed experiments do not prove hardness.

Concrete feasibility includes the whole construction

Quantity	Required accounting
Security	Best attacks, success probabilities and reduction losses
Communication	All public keys and ciphertexts after amplification
Correctness	Proved failure bounds, including exceptional keys
Runtime and memory	Sampling, encryption, decoding and rejection costs

First decision point: does a useful parameter window exist?

A staged first-year programme



Planning estimate. Review feasibility near month 3. Later stages depend on the evidence.

First-quarter deliverables

01 **An assumption and reduction ledger**

Every sampler, task, parameter conversion and security claim.

02 **A reproducible baseline and attack suite**

Exact small-instance checks, benchmarks and attack outcomes.

03 **A parameter report and continuation decision**

Full cost estimates, uncertainty and the bottleneck worth studying.

Focused construction improvements

Candidate direction	Evidence needed
A better trapdoor sampler	A proof for the exact public-key distribution
Tighter reduction constants	A concrete improvement in the viable parameter range
Cheaper amplification	A complete privacy and correctness argument
More efficient encoding	A joint analysis of relation leakage and correlated errors

Choose one direction after the baseline identifies the dominant cost.

The raw decoder exposes its secret to an oracle

$$\text{Dec}(\mathbf{c}) = \mathbf{s}^T \mathbf{c}$$

$$\text{Dec}(\mathbf{u}_i) = \mathbf{s}_i$$

m unit-vector queries
recover the secret support.

Separate construction work

Strong CPA security

Ciphertext validation

A proved CCA conversion

Failure-safe decapsulation

This is a limitation of the raw bit scheme. ABW's existence theorem targets passive security.

A concrete first project

Concrete security and feasibility of sparse-equation public-key encryption

Proceed when

A defensible parameter window survives.
One bottleneck has a plausible remedy.

Redirect when

An attack invalidates the candidate.
Full costs overwhelm the intended benefit.

Suggested team: a proof lead and an attack/implementation lead, with independent review.

Appendix: parameter and proof distinctions

Issue	Precise interpretation
3-LIN-only regime	$m = \Theta(n^{1.4})$, $\varepsilon = \Theta(n^{-0.2})$, with suitable constants
Public-key overlap	A constant overlap bound, not negligible distance
d-LIN + DUE conversion	Public locality D versus search locality $D/2$
Noise conversion	Ciphertext $\varepsilon = 2\eta(1 - \eta)$ for the paired construction
Sample count	Public-key rows differ from reduction sample needs

Appendix: exact raw failure probability

$$E[(-1)^{(\sum e_i)}] = (1 - 2\varepsilon)^r$$

$$\text{Pr}[\text{odd}] = [1 - (1 - 2\varepsilon)^r] / 2$$

Illustration: $\varepsilon = 0.01$	$r = 4$	$r = 16$	$r = 64$
Raw failure probability	3.88%	13.81%	36.28%

The relevant noise is the parity on the secret relation, not the total number of errors.

Appendix: requirements for a complete KEM

Layer	Proof or implementation obligation
Base encryption	Message space, correctness and full CPA security
CCA transformation	Reencryption, validation, rejection and proof losses
Quantum analysis	Base hardness plus a valid quantum reduction chain
Implementation	Constant-time code, randomness and input handling
Protocol integration	Key authentication, derivation and authenticated encryption

Selected references: foundations

Applebaum, Barak & Wigderson

Public-Key Cryptography from Different Assumptions (2009 / 2010)

Regev

On Lattices, Learning with Errors, Random Linear Codes, and Cryptography

Langlois & Stehlé

Worst-Case to Average-Case Reductions for Module Lattices

Yu & Zhang

Public-Key Encryption from Constant-Noise LPN (CRYPTO 2016)

NIST FIPS 203

Module-Lattice-Based Key-Encapsulation Mechanism Standard (2024)

Holenstein & Renner

One-Way Secret-Key Agreement and Applications to Circuit Polarization and Immunization (2005)

Selected references: current research

Bogdanov, Kothari & Rosen (2023)

Public-Key Encryption, Local Pseudorandom Generators, and the Low-Degree Method

Dao & Jain (2025)

Lossy Cryptography from Code-Based Assumptions: Dense-Sparse LPN

Hair & Sahai (April 2026 preprint)

Public Key Encryption from High-Corruption Constraint Satisfaction Problems

HQC team (22 August 2025)

HQC specification and NIST selection status

Hofheinz, Hövelmanns & Kiltz (2017)

A Modular Analysis of the Fujisaki–Okamoto Transformation

NIST SP 800-227 (2025)

Recommendations for Key-Encapsulation Mechanisms